

**DATA VORTEX A'26
ROUND 2
REBUILDING THE SOCIAL ENGINE**

NLP MODEL — TECHNICAL REPORT

Generated from validated local evaluation run
2026-09-18 UTC

1. PROBLEM DEFINITION

The project rebuilds a Social Engine's semantic-understanding layer by classifying social posts. The primary task is sentiment classification (sentiment_label); the secondary task is topic classification (topic_category). Separate models are used rather than a synthetic multi-task architecture, because each target has its own label space and independent validation evidence.

2. DATASET UNDERSTANDING

Dimensions: 9,000 rows × 4 columns. Columns: text_id, post_text, sentiment_label, topic_category. Missing values: {'text_id': 0, 'post_text': 0, 'sentiment_label': 0, 'topic_category': 0}. Exact duplicate rows: 0; normalized duplicate-text rows: 2087 in 987 repeated groups. Duplicate-label consistency: {'sentiment_label': {'conflicting_duplicate_groups': 0, 'conflicting_duplicate_rows': 0}, 'topic_category': {'conflicting_duplicate_groups': 0, 'conflicting_duplicate_rows': 0}}.

3. PREPROCESSING PIPELINE

Raw post_text → quality audit → duplicate-only NFKC/lowercase/whitespace normalization → group-aware split → TF-IDF word 1–2 grams → linear classifier. URLs, mentions, hashtags, punctuation, casing and emoji are retained in the raw text given to TF-IDF; this avoids prematurely stripping sentiment and topical signals. The vectorizer is fit inside each training pipeline only.

4. MODEL SELECTION

A majority baseline establishes a minimal reference. TF-IDF + Logistic Regression provides probabilistic, interpretable linear classification; TF-IDF + Linear SVM is a strong sparse-text classifier. Macro F1 on validation chose the final candidate, balancing class performance rather than optimizing accuracy alone.

sentiment_label: model comparison

Model	Accuracy	Macro F1	Weighted F1
TF-IDF + Logistic Regression	0.5809	0.5763	0.5833
TF-IDF + Linear SVM	0.5708	0.5670	0.5736
Majority baseline	0.2876	0.1489	0.1285

topic_category: model comparison

Model	Accuracy	Macro F1	Weighted F1
TF-IDF + Linear SVM	0.9146	0.5970	0.8951
TF-IDF + Logistic Regression	0.9011	0.5842	0.8872
Majority baseline	0.8494	0.2296	0.7803

5. TRAINING METHODOLOGY

Random seed: 42. A GroupShuffleSplit isolates 20% of normalized-text groups as test data. A second group split assigns 12.5% of the remaining groups to validation, yielding approximately 70/10/20 train/validation/test. Group intersections were verified as zero. After validation selection, the winning exact pipeline was retrained on train+validation and evaluated once on test.

6. EVALUATION METRICS

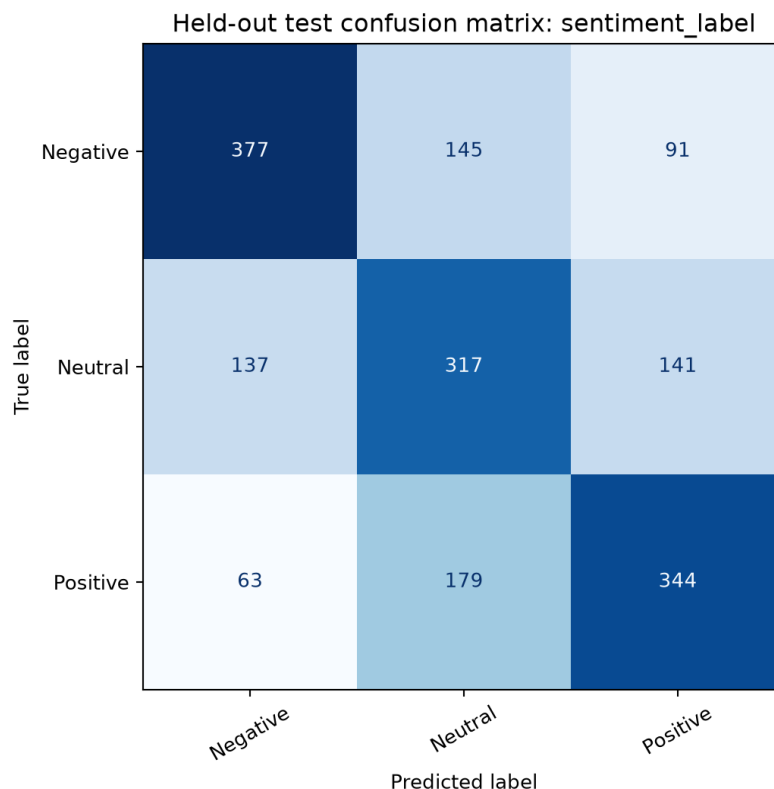
Accuracy gives overall correctness. Macro precision, recall and F1 treat each class equally; weighted F1 reflects support-weighted performance. Per-class report values and supports appear in the metrics report and JSON results.

sentiment_label: held-out test metrics

Final model: TF-IDF + Logistic Regression.

Metric	Value
Accuracy	0.5786
Macro precision	0.5817
Macro recall	0.5783
Macro F1	0.5795
Weighted F1	0.5800

7. CONFUSION MATRIX



Rows are actual labels and columns are model predictions.

8. ERROR ANALYSIS

Observed confusion pairs: Positive → Neutral (179); Negative → Neutral (145); Neutral → Positive (141); Neutral → Negative (137); Negative → Positive (91). 756 errors were identified from all held-out predictions. Common limitations include short/noisy posts, ambiguous tone, sarcasm that sparse lexical features cannot reliably infer, and semantic overlap between topic labels. Example errors are shown below without adding information beyond the supplied text.

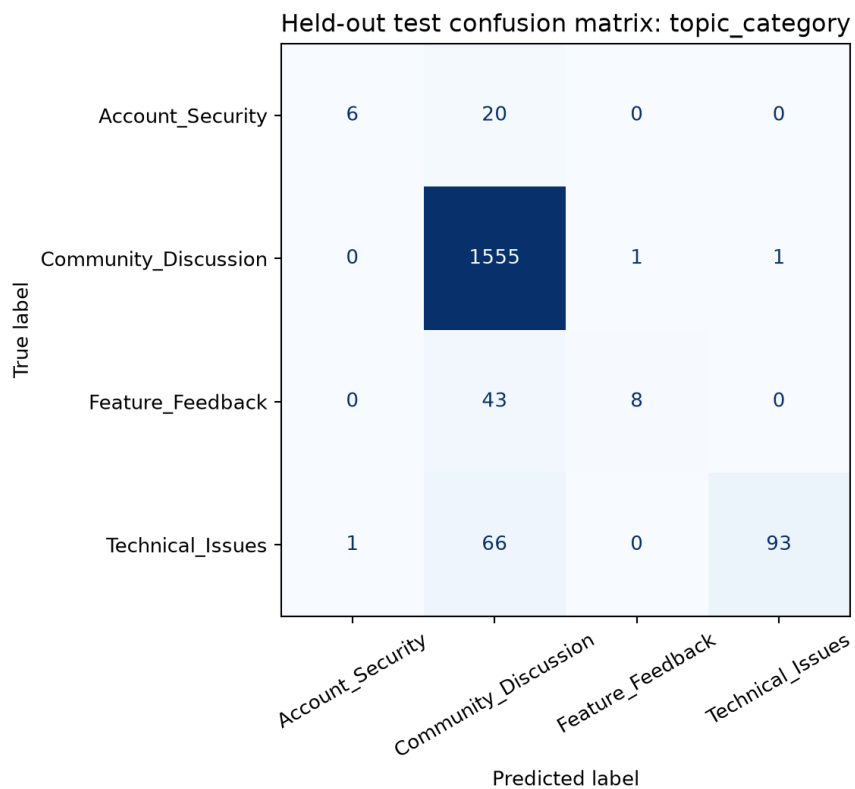
Actual	Predicted	Post excerpt
Negative	Positive	@user @user aaaah. Nokia used to make the Best Phone Cameras Ever. Sadly I think those days may b
Neutral	Negative	"B4 d hood was placed on Yakub's face, he insisted that he wanted 2 c d rising sun, which he was not all
Positive	Negative	In 7th grade I was introduced to the book Twilight and that's the only book that kept me out of my world.

topic_category: held-out test metrics

Final model: TF-IDF + Linear SVM.

Metric	Value
Accuracy	0.9264
Macro precision	0.9147
Macro recall	0.4919
Macro F1	0.5805
Weighted F1	0.9110

7. CONFUSION MATRIX



Rows are actual labels and columns are model predictions.

8. ERROR ANALYSIS

Observed confusion pairs: Technical_Issues → Community_Discussion (66); Feature_Feedback → Community_Discussion (43); Account_Security → Community_Discussion (20); Community_Discussion → Feature_Feedback (1); Community_Discussion → Technical_Issues (1). 132 errors were identified from all held-out predictions. Common limitations include short/noisy posts, ambiguous tone, sarcasm that sparse lexical features cannot reliably infer, and semantic overlap between topic labels. Example errors are shown below without adding information beyond the supplied text.

Actual	Predicted	Post excerpt
Technical_Issues	Community_Discussion	Apparently Justin Bieber is in the area. I know this because I have students (plural) missing school tomorrow.
Technical_Issues	Community_Discussion	Misinterpreting Mariah Carey lyrics. Apparently she wants to fly a family into the sun and something about
Account_Security	Community_Discussion	David Miliband on #BBCQT! This is going to be a wild Thursday night!

9. RESULTS & FINDINGS

Results are strictly held-out measurements produced by the included run script. The saved full pipelines can predict new post text without separately reconstructing vectorization.

10. LIMITATIONS

This is a labelled, finite dataset. Duplicate text was handled safely but its prevalence may limit independence of source content. Sparse linear models do not fully capture irony, context, or unseen vocabulary, and performance should not be interpreted as guaranteed generalization beyond this dataset.

11. REPRODUCIBILITY

Python 3.14.0; seed 42; relative dataset path data/Labeled_Social_NLP_Training_Data.csv. Run python run_round2.py from repository root. Dependencies are pinned in requirements.txt. Complete pipelines are models/*_pipeline.pkl.

12. CONCLUSION

A reproducible, leakage-safe two-task NLP submission was built with validation-based model selection, untouched held-out evaluation, generated confusion matrices, structured error artifacts, and aligned reports.